Soft Quantum Kernels for Workforce Risk Analytics

Convergence Guarantees and Operational Cost Analysis

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Outline

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The Most Underestimated Operational Risk

What is the most costly, least measured operational risk in a large organisation?

Unplanned Employee Departures

- Replacing one employee costs **50–200%** of annual salary
- IBM HR benchmark: 1,470 staff, 35 variables, $\approx 16\%$ attrition
- Missing a departure (false negative) $\gg$ a false alarm

Why Standard ML Falls Short

- HR data is **nonlinear and high-dimensional**: overtime $\times$ salary $\times$ career stage $\times$ team dynamics
- Classical models miss **higher-order interactions**
- Quantum methods can capture them but quantum hardware is noisy and inaccessible

5,000-Person Organisation

$50K to replace vs. $5K to retain. **Cost ratio 10:1 against inaction.**

Our answer: *Soft quantum kernels*
quantum-inspired maths on ordinary GPUs.

From Classical to Quantum Kernels: A Practitioner's Intuition

Classical Kernel (SVM-RBF)

A kernel is a **similarity score** between two data points.

RBF sees only **pairwise distances**.
It cannot capture three-way or four-way HR interactions natively.

The Interaction Problem

Modelling overtime $\times$ satisfaction $\times$ tenure $\times$ distance simultaneously requires an exponentially large feature space prohibitive for classical methods.

Quantum Kernel Idea

Map each employee's vector x to a quantum state $|\psi(\mathrm{x})\rangle$ in a high-dimensional Hilbert space:

$$k(\mathrm{x}_i, \mathrm{x}_j) = |\langle\psi(\mathrm{x}_i)|\psi(\mathrm{x}_j)\rangle|^2$$

The inner product $\langle\cdot|\cdot\rangle$ captures all higher-order interactions implicitly.

Key intuition: entanglement between qubits = interaction between HR variables.
Parameter τ controls how much.

The Soft Innovation

We **relax** the discrete quantum circuit into a smooth function computable on a GPU in $\sim$45 min, noise-free, no quantum access needed.

The Core Construction: Soft Unitary Operator

From Discrete Circuits to a Continuous Path

A standard quantum circuit applies gates sequentially: encode, rotate, entangle, repeat.

Problem: non-differentiable; requires quantum hardware.

Our idea: merge all layers into one smooth exponential:

$$U_\tau(\mathrm{x}, \boldsymbol{\theta}) = \exp\Big(-i \sum_{\ell=1}^{L} \big[H_{\mathrm{enc}} + H_{\mathrm{rot}} + \underbrace{\tau H_{\mathrm{ent}}}_{\text{dial}}\big]\Big)$$

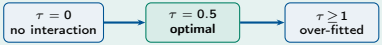
$\tau > 0$: **relaxation temperature**, single knob for complexity.

The Soft Kernel

$$k_{\boldsymbol{\theta}}^{(\tau)}(\mathrm{x}_i, \mathrm{x}_j) = \left|\langle 0|^{\otimes n} U_\tau(\mathrm{x}_i)^\dagger U_\tau(\mathrm{x}_j)|0\rangle^{\otimes n}\right|^2$$

PSD · normalised · C^∞-smooth · recovers ideal kernel as $\tau \to 0$.

The Temperature Dial τ



- Too low: model misses patterns
- Too high: overfits, generalisation drops
- $\tau^* = 0.5$ found by cross-validation

Efficient GPU Computation

- Cost $\mathcal{O}(2^{3n})$ via Padé; $n \leq 12$ on commodity GPUs
- **Exact, differentiable, noise-free**

Why Quantum Parameters Live on a Curved Surface

The Unitarity Constraint: $U^\dagger U = I$

$\dagger$ = conjugate transpose. A unitary operator is a *generalised rotation*: every vector keeps its length:

$$\|Uv\| = \|v\| \quad \forall v$$

This is a **physical law**: quantum mechanics requires total probability to be preserved.

What Breaks if $U^\dagger U \neq I$

- Probabilities no longer sum to 1 $\rightarrow$ **physically meaningless**
- Kernel loses positive semi-definiteness $\rightarrow$ SVM invalid
- Predictions degrade with no error bound

The Stiefel Manifold $\mathrm{St}_{\mathbb{C}}(k, 2^n)$

$$\mathrm{St}_{\mathbb{C}}(k, 2^n) = \left\{ V \in \mathbb{C}^{2^n \times k} : V^\dagger V = I_k \right\}$$

The **set of all valid parameter matrices** a smooth curved surface inside $\mathbb{C}^{2^n \times k}$. When $k = 2^n$: reduces to $U(2^n)$.

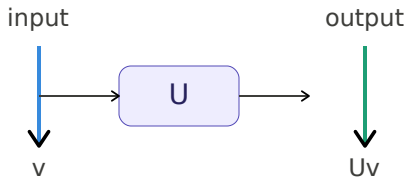
Key Geometric Objects

- **Tangent space** at V:
  $T_V\mathrm{St} = \{\Delta : V^\dagger \Delta + \Delta^\dagger V = 0\}$
  directions of valid infinitesimal moves
- **Riemannian metric**: $\langle \Delta_1, \Delta_2 \rangle_V = \mathrm{Re}\,\mathrm{Tr}(\Delta_1^\dagger \Delta_2)$
  measures angles and lengths on the surface

Why quantum parameters live on a curved surface

The unitarity constraint: $U^\dagger U = I$

A unitary operator U preserves length



$\|v\| = \|Uv\|$ (same length!)

If we break unitarity…

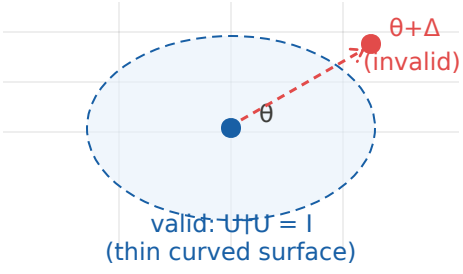
quantum state probabilities no longer sum to 1 → physically meaningless

kernel loses positive semi-definiteness → SVM has no valid solution

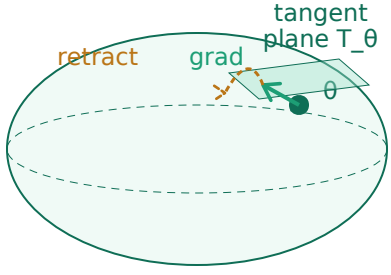
predictions degrade unpredictably, no error bound possible

The Stiefel manifold: the space where valid parameters live

Flat parameter space $\mathbb{R}^p$
(what gradient descent assumes)



Stiefel manifold $St(k, 2^n)$
(curved surface of unitary operators)



The QR retraction: how each training step works

Start
θ_t on manifold
$U^\dagger U = I$ ✓

→

Riemannian grad
project Euclidean grad to T_θ

→

Raw update
$\theta_t - \eta \cdot \text{grad}$
leaves manifold ✗

→

QR retraction
factorise → keep Q
θ_{t+1}, $U^\dagger U = I$ ✓

What QR factorisation does

ake the (slightly invalid) updated matrix $V = \theta_t - \eta \cdot \text{grad}$

actorise: $V = Q \cdot R$, where Q has orthonormal columns ($Q^\dagger Q = I$) and R is upper-triangular

eep only Q — it is by construction unitary. Set $\theta_{t+1} = Q$

Why this gives certified convergence

very iterate is valid. The loss decreases by $\geq \|\text{grad}\|^2/2\beta_s$ per step (descent lemma on manifolds) → $O(\kappa_s \log 1/\varepsilon)$ s

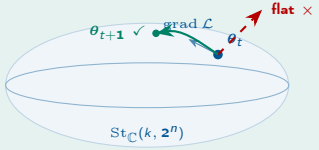
Riemannian Gradient Descent and the QR Retraction

Why Flat Optimisation Fails

Standard GD: $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla \mathcal{L}$

On a curved surface this step **leaves the manifold**: $\boldsymbol{\theta}_{t+1}$ is no longer unitary. Re-projecting is costly and destroys convergence guarantees.

Riemannian GD: Stay on the Surface



The QR Retraction: Four Steps

1. **Riemannian gradient**
   $\operatorname{grad} \mathcal{L} = \nabla \mathcal{L} - \boldsymbol{\theta}_t (\nabla \mathcal{L})^\dagger \boldsymbol{\theta}_t$
2. **Raw step** (leaves manifold)
   $V = \boldsymbol{\theta}_t - \eta \cdot \operatorname{grad} \mathcal{L}, \quad V^\dagger V \neq I$
3. **QR factorisation**
   $V = QR: \; Q^\dagger Q = I_k, \; R$ upper-triangular
4. **Retract**: $\boldsymbol{\theta}_{t+1} = Q$

Why QR?

- Q is **exactly unitary** by construction zero approximation error
- Numerically stable, **parallelisable on GPU**
- Faster than SVD-based retraction in practice
- Every iterate: $U^\dagger U = I$ **certified** $\rightarrow$ auditable, deployable

The Convergence Theorem: what does It Means for a Risk Manager?

Theorem 1 (informal): Certified Convergence on the Stiefel Manifold

Starting from any parameters, Riemannian GD reaches an ε-optimal solution in at most

$$\mathcal{O}\big(\kappa_s \cdot \log(1/\varepsilon)\big) \quad \text{iterations,}$$

where the **condition number** decomposes into three interpretable factors:

$$\kappa_s \;\leq\; \underbrace{(1+\tau L C_{\mathrm{curv}})}_{\substack{\text{soft relaxation}\\ \leq 2.5\times}} \cdot \underbrace{w_{\max}/w_{\min}}_{\substack{\text{cost asymmetry}\\ \approx 5\times}} \cdot \underbrace{\kappa}_{\substack{\text{intrinsic}\\ \text{difficulty}}} \quad \Rightarrow \quad \text{total overhead} \;\approx 12.5\times$$

Reading the Formula

- $\log(1/\varepsilon)$ grows very slowly: doubling accuracy adds only ≈ 3 extra iterations
- $12.5\times$ overhead is **modest** fully recovered by recall gains
- **No quantum hardware** at any step

Why Certification Matters Operationally

Without guarantees, a model is a black box.

This theorem certifies quality bounds: if training completes, the solution is auditable essential for DORA/Basel IV contexts.

A Structural Fit: HR Data and Quantum Methods

Theorem 2 (Soft Separation Bounds)

The soft kernel margin **strictly exceeds** any classical kernel:

$$\gamma_{\text{Soft-QFE}} \geq \gamma_{\text{classical}} + \Omega\left(\sqrt{d_b/d}\right) - \mathcal{O}(\tau L(n-1))$$

IBM: d_b=3 of d'=8 $\Rightarrow$ structural advantage $\approx$ 0.61.

Key Thesis

Quantum management science is **not a transplantation** of physics algorithms.

Organisational data, mixed types, binary flags, asymmetric costs, multi-party governance maps *naturally* onto quantum-geometric methods.

Four Natural Alignments

1. **Binary features** (overtime, travel) give exact π-rotations, max discrimination, zero error
2. **Asymmetric costs** (departure $\gg$ false alarm) are *native*
3. **Multi-stakeholder**: one AUC, different thresholds for HR/Finance/Ops
4. **High-order interactions** (satisfaction $\times$ pay $\times$ career) native to quantum feature spaces

Handling Class Imbalance: Entropy-Regularised Nyström

The Imbalance Problem

Only 16% of employees leave. Standard Nyström selects landmarks uniformly: 84% come from stayers, so the minority class is under-represented and **recall degrades**.

Entropy-Regularised Landmark Selection

Choose m landmarks by solving:

$$\min_{\boldsymbol{p}} \underbrace{\|K - \tilde{K}\|_F^2}_{\text{approx. error}} - \underbrace{\lambda \sum_i p_i \log p_i}_{\text{spread landmarks}} + \underbrace{\mu\, D_{\mathrm{KL}}(\boldsymbol{p} \| \boldsymbol{q}_{\mathrm{strat}})}_{\text{boost minority}}$$

$q_i \propto \max(\pi_{y_i}, \tau_{\mathrm{strat}})$: deliberately over-samples leavers.

Guarantees

- **Differentiable**: $\boldsymbol{p}^*(\boldsymbol{\theta}, \tau)$ is C^∞ entire pipeline remains end-to-end trainable
- **Approximation**: $\mathbb{E}[\|K - \tilde{K}\|_F] \leq \varepsilon \|K\|_F$
- **Minority fidelity** improves *exponentially* vs. uniform sampling
- Only $m = 200$ landmarks needed (from 1,470 training points)

Experimental Setup

Dataset: IBM HR Analytics

- 1,470 employees, 35 variables, attrition $\approx 16\%$
- 8 features: OverTime, MonthlyIncome, Age, TotalWorkingYears, JobSatisfaction, YearsAtCompany, DistanceFromHome, NumCompaniesWorked
- 5-fold stratified nested CV
- Best: τ=0.5, C=10, w_+/w_-=5

Soft-QC Pipeline

- n=4 qubits, L=3 layers, m=200 landmarks
- PyTorch, Pymanopt, scikit-learn
- **No quantum hardware**
- Total: $\sim$ 45 **min on NVIDIA A100**

Results: State-of-the-Art on Every Metric

Minority-class metrics, IBM HR Analytics benchmark

Table: *

Method	Acc.	Prec.	Recall	F1	AUC
SVM-RBF (classical)	0.857	0.667	0.500	0.571	0.820
Random Forest	0.864	0.700	0.583	0.636	0.840
XGBoost	0.881	**0.750**	0.563	0.643	0.850
Q-SVM (discrete)	0.878	0.714	0.625	0.667	0.858
Soft-QC (ours)	**0.884**	0.735	**0.652**	**0.691**	**0.871**

What the Numbers Mean

- **Recall 0.652**: flags 65% of future leavers, the metric that saves money
- **AUC 0.871**: one model, three stakeholders (HR/Finance/Ops), no retraining
- Generalisation gap: 1.91% vs. 12.85% (RF)

Speed Advantage

Soft-QC: ∼45 min on GPU
Discrete Q-SVM: ∼4h on simulator
⇒ **5.3× faster**, no quantum HW.

Ablation: Incremental F1 Gains

Component added	ΔF1
Soft kernel	+7.94 pp
Riemannian opt.	+1.59 pp
Entropy Nyström	+1.60 pp
Cost weighting	+0.87 pp

From Performance Metrics to Savings

Scenario: 5,000 employees · 16% attrition · $50K replacement · $5K intervention · 40% success

	SVM-RBF	XGBoost	Q-SVM	Soft-QC
At-risk identified	400	450	500	**522**
Departures prevented (40%)	160	180	200	**209**
Replacement costs saved ($M)	8.0	9.0	10.0	**10.45**
Intervention costs ($M)	2.0	2.25	2.5	2.61
Net savings ($M)	**6.0**	**6.75**	**7.5**	**7.84**

vs. Classical SVM-RBF

$1.84M additional annual net savings

Deployable on existing GPU infrastructure.

vs. Discrete Q-SVM

$340K additional savings

+ no quantum hardware cost

5.3× faster. Fully auditable.

Limitations and Future Directions

Current Limitations

- **Scalability**: cost $\mathcal{O}(2^{3n})$ limits to $n \leq 12$ qubits; Krylov methods extend this but are not yet integrated
- **Dataset**: IBM HR is a well-known but synthetic benchmark; real-world validation is the next step
- **Assumptions**: convergence relies on Polyak–Łojasiewicz and bounded curvature standard but not always verifiable a priori
- **No NISQ comparison**: benchmarked vs. simulator, not real quantum hardware

Immediate Extensions

- Multi-class attrition risk tiers (low/medium/high)
- Temporal kernels for career-trajectory signals
- Federated deployment across business units
- Financial risk, credit scoring, supply-chain default

From Prediction to Prescription

Full differentiability of the soft kernel enables embedding into **two-stage stochastic programmes** for workforce sizing from *who will leave* to *what to do about it*.

Conclusion

Five Contributions

1. **Soft quantum kernel**: quantum-level expressivity on standard GPUs, no quantum hardware, no shot noise
2. **Convergence theorem**: explicit, interpretable condition number training is certified, not hoped for
3. **Separation result**: HR binary features give a τ-independent structural advantage over classical kernels
4. **Entropy Nyström**: differentiable, minority-preserving approximation with exponential fidelity gain
5. **SOTA results**: AUC 0.871, F1 0.691, \$1.84M additional savings, 5.3$\times$ faster

Broader Vision

Soft quantum computing is a **bridge** between the promise of quantum ML and the constraints of organisational practice removing hardware barriers and making quantum-geometric analytics available **today**. The framework extends to financial risk, supply chain, portfolio allocation and public administration.

Thank you! Questions?

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Backup: Formal Theorem Statements

Theorem 1 (Convergence)

Under geodesic smoothness (A1), manifold PL (A2), bounded curvature (A3), Riemannian GD with $\eta = 1/\beta_s$ and QR retraction satisfies:

$$\mathcal{L}_s(\boldsymbol{\theta}_t) - \mathcal{L}_s^* \leq \left(1 - \frac{\mu_s}{\beta_s}\right)^t \left(\mathcal{L}_s(\boldsymbol{\theta}_0) - \mathcal{L}_s^*\right)$$

ε-optimality in $\mathcal{O}(\kappa_s \log(1/\varepsilon))$ steps; $\kappa_s \leq (1 + \tau L C_{\mathrm{curv}}) \cdot \frac{w_{\max}}{w_{\min}} \cdot \kappa$.

Theorem 2 (Soft Separation)

With d_b binary features encoded as $\{0, \pi\}$ and fidelity $F_\tau \geq 1 - \delta_{\mathrm{relax}}$:

$$\gamma_{\text{Soft-QFE}} \geq \gamma_{\mathrm{classical}} + \Omega\left(\sqrt{d_b/d}\right) - \mathcal{O}(\tau L(n-1))$$

Binary features contribute a τ-independent advantage because π-rotation is its own generator.

Backup: Research Programme and References

Quantum Management Science Programme (Ferrara et al.)

1. **Ferrara, Sáez-Ortuño, Forgas-Coll (2026), *Sci. Reports***: First quantum kernel framework for marketing analytics
2. **Sáez-Ortuño et al. (2025), *Ann. Oper. Res.***: Discrete Q-kernels for employee attrition
3. **This work**: Soft quantum kernels, Riemannian convergence, entropy Nyström, GPU deployment

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